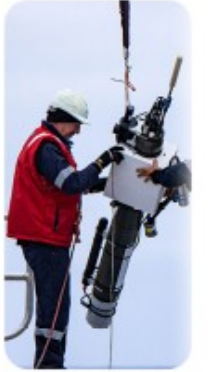


Where can I access IMOS data?

July 2025

Mark Rehbein, Benedicte Pasquer & Victoria Isaac

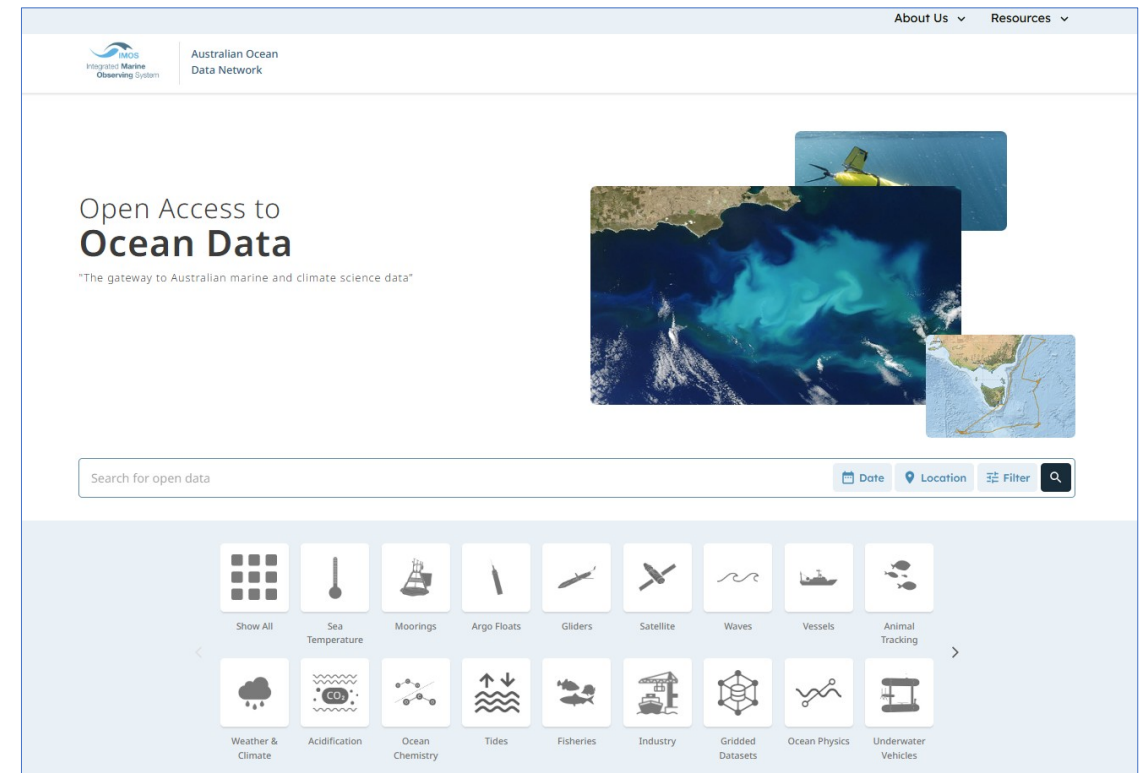
Australian Ocean Data Network (AODN), Integrated Marine Observing System (IMOS)



IMOS acknowledges the Traditional Custodians and Elders of the land and sea on which we work and observe, and recognise them as Australia's first marine scientists and carers of Sea Country. We pay our respects to Aboriginal and Torres Strait Islander peoples past and present.

Outline

1. Overview of the Australian Ocean Data Network (AODN) and Summary of IMOS Data Access Methods – Mark (10min)
2. Accessing Data using the AODN Portal - Bene (10min)
3. Preview of the New AODN Portal - Vicky (10min)



IMOS Data – Access to free & open data via the AODN

Australia Ocean Data Network (AODN): Access to IMOS and other data

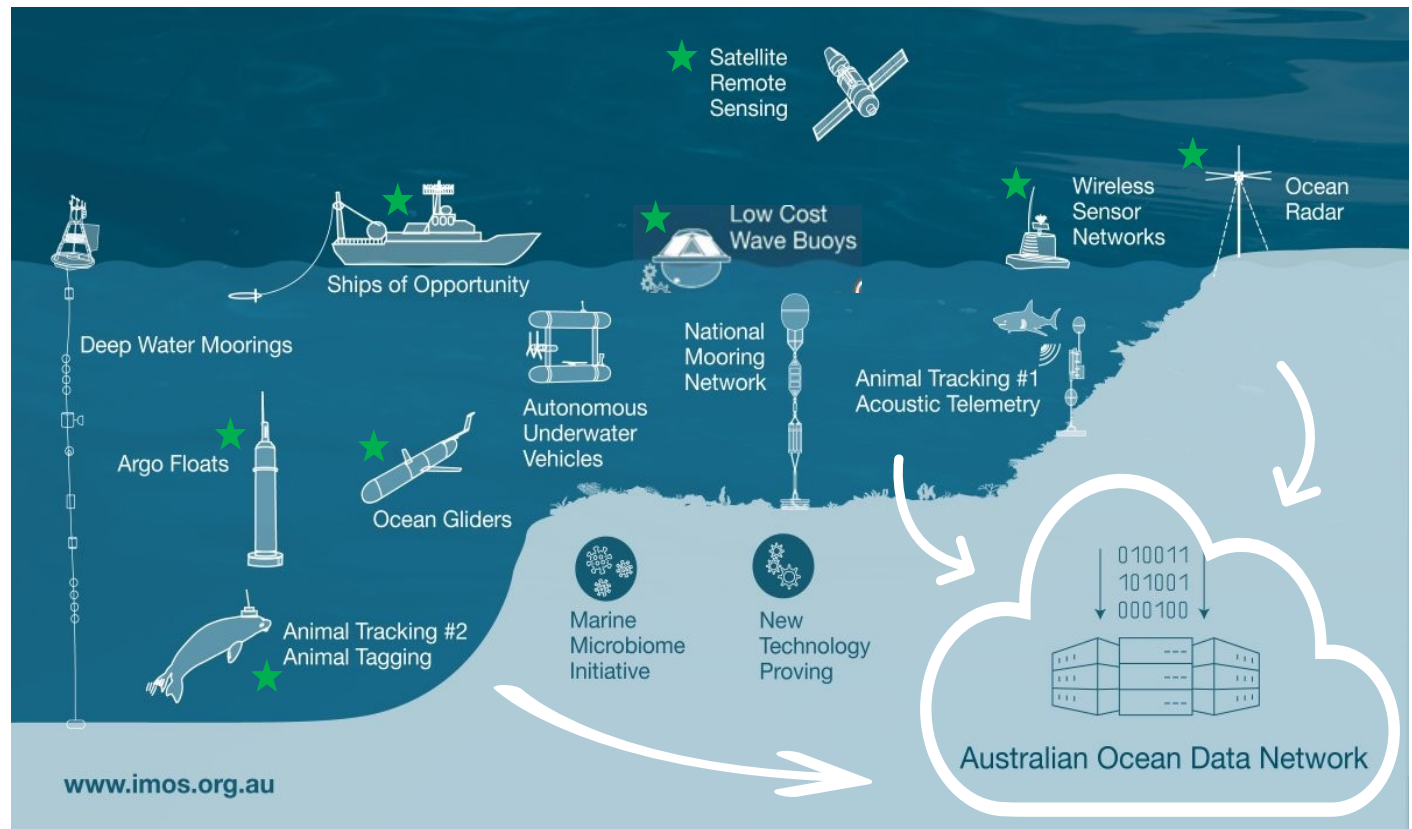
- AODN is the data management component of IMOS
- IMOS data are free and open
- Using the data simply requires attribution/citation

Data accessible via:

- Web applications
- Data services
- Formats

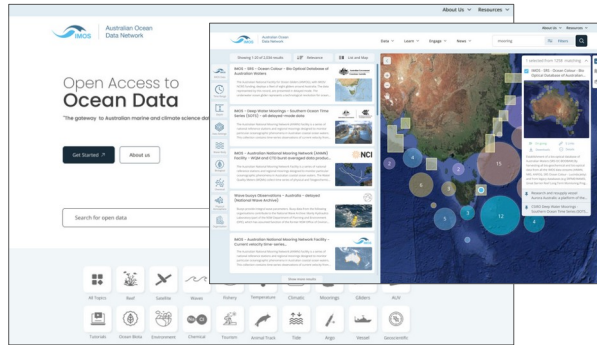
Targets different user groups, with different tools, skills and capabilities

Goal: to maximise the use and uptake of data



★ = near real time component

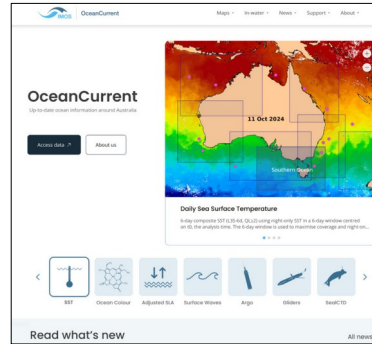
Access methods for less technical use cases



IMOS AODN Portal

- Focus on data discovery
- Interactive experience
- Data previews
- Subset & download (CSV/Excel)
- New version under development

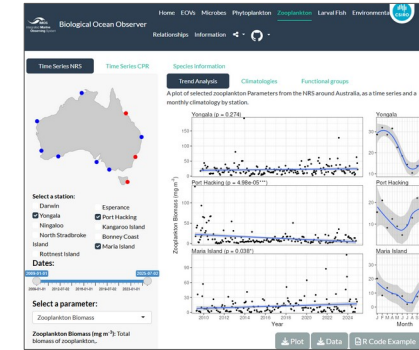
Bene & Vicky



IMOS Ocean Current

- Focus on physical oceanographic
- Visualisations
- Remotely sensed obs as maps
 - Sea level
 - Ocean currents
 - SST
 - Surface waves
- Plus in water observations
 - Argo, gliders
 - CTD's, moorings
- At local, state and national scales
- New version under development

Gabi



IMOS Biological Ocean Observer

- Focus on biological data
 - Microbial
 - Phytoplankton
 - Zooplankton
 - Larval fish
 - Biogeochemistry
- Interactive experience
- Visualisations (trends)

Claire

Access methods for GIS and forecasting use cases



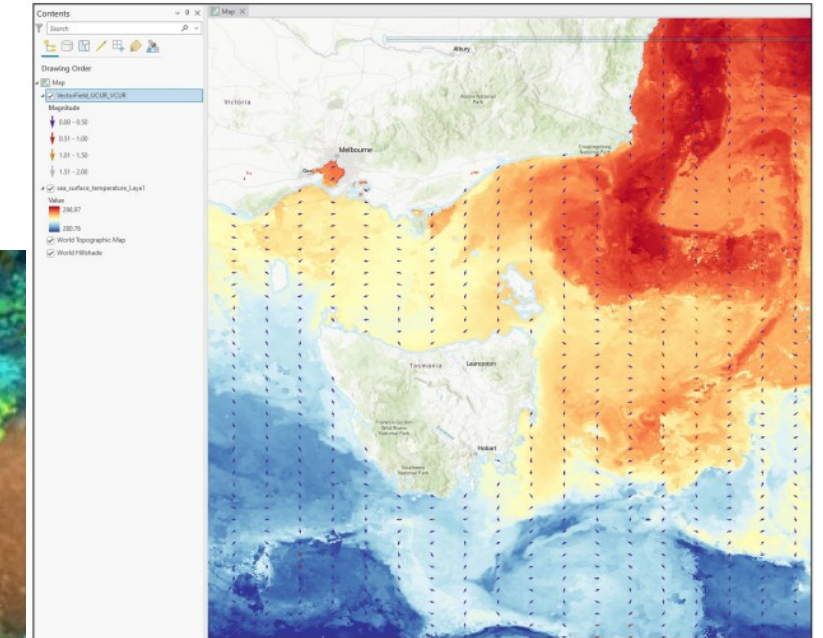
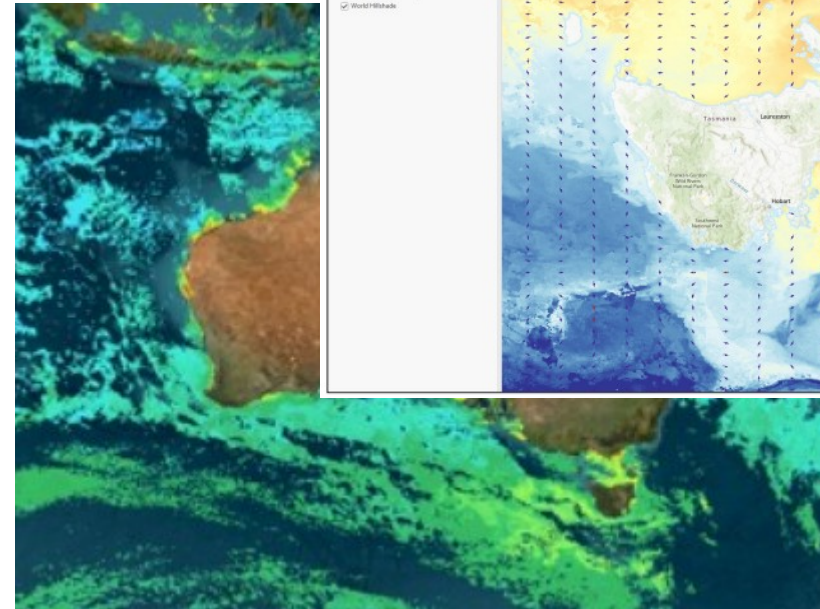
GIS / Spatial Analysis Tools

- GIS users
- Industry & Government
- ArcGIS / QGIS
- Uses AODN spatial web services (WMS/WFS)

Ben

Weather Forecasting & Climate Modelling

- Via the WMO data delivery systems (GTS/WIS2)



Visualisation in QGIS

Access methods for more technical use cases



THREDDS / NetCDF/ OpenDAP

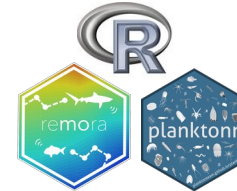
- Oceanographers & Modellers
- Most IMOS data are available in this format
- Gridded & multi-dimensional datasets



Cloud Optimised Formats

- Data science use-cases
- Rapid, robust extractions from large datasets

Eduardo



R Packages

- Data science use-cases
- remora (R)
- planktonr (R)
- Vignettes available



AODN Data Science Notebooks

- Mostly in Python/Jupyter
- Cloud optimised & THREDDS

Bene

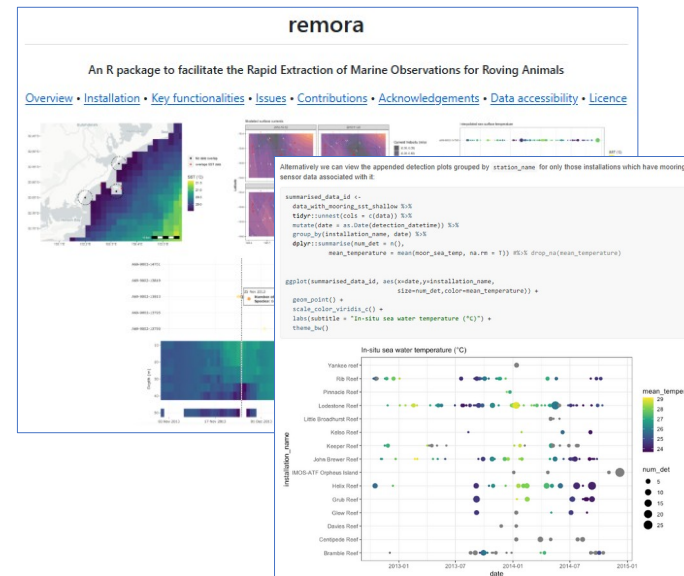
For data experts with skills in data analysis tools:

- Matlab, Python, R

To do data intensive things like:

- Data analysis and exploration
- Advanced or adhoc visualisation
- AI/ML, modelling etc

Eduardo & Denisse – R tutorials



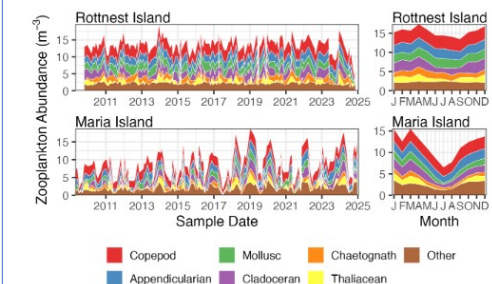
Functional Groups

We can also gain a lot of insight into the community structure by looking at how the composition (or proportions) of the functional groups changes over time, long term and seasonally. In this function we have chosen some of the more important function groups and plotted them as a time series and a seasonal cycle. Again you can see that Maria Island demonstrates a strong seasonal pattern with lower abundances in all groups during winter.

```
FgZ <- pr_get_FunctionGroups(survey = "NRS", Type = "Zooplankton") %>%
  dplyr::filter(StationCode %in% c("ROT", "MAL"))

p1 <- planktonr::pr_plot_tsfg(FgZ, Scale = "Actual")
p2 <- planktonr::pr_plot_tsfg(FgZ, Scale = "Actual", Trend = "Month") +
  ggplot2::theme(axis.title.y = element_blank())

p1 + p2 +
  patchwork::p1plot_layout(widths = c(3,1), guides = "collect") &
  theme(legend.position = "bottom")
```



remora and planktonr R vignettes

Questions?

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Integrated Marine Observing System



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Bene Pasquer

Product Manager, Australian Ocean Data Network
Integrated Marine Observing System



<https://imos.org.au/>

<https://portal.aodn.org.au/>

<https://oceancurrent.aodn.org.au/>

<https://shiny.csiro.au/BioOceanObserver/>

